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Lesson 1: Bayesian Probability
Lesson 2: Bayesian Inference
Reading: Bayesian Reading 2h
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Practice Assignment: Bayesian Inference Quiz Submitted
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Module 1 Summary

Your grade: 100%

Your latest: **100%** • Your highest score: **100%** • Let's keep learning! To pass you need at least 80%. We keep your highest score.

Next item

Bayesian Inference Quiz

Instructions

Review Learning Objectives

coach
Ready to review what you've learned before starting the assignment? I'm here to help.
Help me practice Let's chat

Assignment details

Submitted **Attempts**

Your grade
100%

Try again

View submission 1 attempt

Feedback

1. Which of these comes first in the process of computing the posterior distribution in Bayesian inference? **1 / 1 point**

- (•) Specifying a prior distribution
- () Applying Bayes' rule
- () Computing the likelihood
- () Normalizing the posterior distribution

Correct
Correct! Once we have specified a prior distribution, we can use data collected to assess and construct the posterior. But the prior distribution is needed before anything can be done.

2. In Bayesian inference, what does the likelihood function represent? **1 / 1 point**

- (•) The probability of observing the data given the prior distribution
- () The updated beliefs reflected in probabilities
- () Marginal probability
- () The probability represented the distribution of our initial beliefs

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Correct
Correct! Given the prior, we can determine the likelihood of the observed data. Based on the likelihood of the data, we adjust our beliefs (compute the posterior from the prior).

3. Suppose you are given a pair of digital calipers that measure lengths in millimeters and you are told that they have not been calibrated correct. They are either generally off by -1 mm or 1 mm, with no reason to believe one outcome is more likely than the other. Digital calipers have errors in their measurements that follow normal distributions. For these calipers, the standard deviation is 1 mm. **1 / 1 point**

Testing these calipers leads to a measurement of 0.5mm larger than what you would expect. In the face of this evidence, what is the probability that the calipers are off by 1 mm?

Hint: Recall that for a normal distribution, we can say

$$p(y|\mu = x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y-x)^2}{2\sigma^2}}$$

- () There is not enough information to calculate this.
- () 0.500
- (•) 0.731
- () 1.000

Correct
Correct! As one would guess, the probability should increase in light of this evidence.